

Mahboob Alam

ML Engineer | Computer Vision | LLM & RAG | Open to Full-Time

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Education

M.Tech in Data Science – CGPA: 8.10/10

2024 – 2026

Defence Institute of Advanced Technology (DIAT-DU), Pune

B.Tech in Computer Science and Engineering

2019 – 2023

RBSSIET, Rewari (IGU Affiliated)

Technical Skills

Languages Python, SQL

ML / DL PyTorch, Scikit-learn, CNNs, Transformers, Semantic Segmentation, Physics-Informed Neural Networks, LightGBM, SHAP, Grad-CAM

LLM & RAG Hugging Face, QLoRA, PEFT, BitsAndBytes, Sentence-Transformers, Qdrant, LangChain, Retrieval-Augmented Generation

Tools FastAPI, Docker, Docker Compose, GitHub Actions (CI/CD), Streamlit, OpenCV, Pandas, NumPy

Research Experience

Research Intern – MoES, Govt. of India Sponsored

May 2025 – April 2026

Defence Institute of Advanced Technology (DIAT-DU), Pune

Pune, India

- Designed **PIAU-Net** by embedding the Jaffe-McGlamery optical scattering model into an Attention U-Net – the first physics-informed architecture for underwater fish segmentation – trained on NVIDIA RTX 6000 Ada (48 GB) with mixed precision; achieved **97.38% mIoU** on LFish and **93.98%** on AquaOV255, outperforming DeepLabV3+ by **+3.44pp**
- Built **LiteFishSeg** (MobileNetV3+BiFPN+FCOS, 9.08M params) achieving **80.3% mIoU** at **2.7x fewer parameters** than YOLOv10l-seg; and **FishSegDet** (ConvNeXt-V2-L+BiFPN+TAL) reaching **70.0% mAP**, **95.1% mAP50**, **99.1% PA** – surpassing YOLOv11l-seg by **+7.10pp mAP50**

Publication

Mahboob Alam, S. V. Dhavale, D. Srikanth. *PIAUNet: An Enhanced U-Net Framework for Underwater Segmentation in Aquaculture.* IJTE, ISTE, Vol. 48, No. 2, Dec 2025.

Projects

Distill – Fine-Tuned Qwen2.5-3B for Text-to-SQL (QLoRA)

Python | PyTorch | QLoRA | PEFT | BitsAndBytes |

FastAPI | Docker

github.com/MahboobAlam0/Distill--Fine-tuned-Qwen2.5-3B

- Fine-tuned Qwen2.5-3B for NL-to-SQL generation on a free Kaggle T4 using QLoRA (4-bit NF4, LoRA r=16/alpha=32) on 28K training examples; built a custom evaluation harness measuring exact match accuracy, BLEU score, and SQLite execution accuracy to simulate real query correctness
- Improved exact-match accuracy by **13x (2.33% to 30.67%)** and reached **55.03% execution accuracy** on 7,858 test examples; shipped a **114 MB LoRA adapter** instead of a 6 GB full model, deployed via FastAPI, Docker, and GitHub Actions CI/CD

AI Clinical RAG – Evidence-Grounded QA over PubMed

Python | Sentence-Transformers | Qdrant | Groq

Llama-3.3-70B | FastAPI | Docker

github.com/MahboobAlam0/ai-clinical-rag

- Fine-tuned all-MiniLM-L6-v2 on 4,500 PubMedQA question-context pairs using MultipleNegativesRankingLoss (10 epochs); indexed 5,355 PubMed abstracts in Qdrant with a cosine-score confidence estimator that flags answers as HIGH, MED, or LOW confidence based on retrieval signal strength
- Improved retrieval nDCG@10 from **0.806 to 0.926 (+12%)** and Accuracy@1 from **0.700 to 0.823**; delivered SSE streaming at 874 ms average latency with LRU caching, per-IP rate limiting, request tracing, and **80% test coverage** enforced in CI

Policy-Aware Credit Decision Engine

Python | LightGBM | SHAP | FastAPI | Streamlit | Docker | GitHub Actions

github.com/MahboobAlam0/policy-aware-credit-decision-engine

- Separated a Platt-scaled LightGBM probability-of-default model from a configurable business policy layer (EL = PD x LGD x EAD) with dual-threshold approve/review/decline logic and fairness controls – replacing a monolithic system where ML risk and business rules were tightly coupled and impossible to audit independently
- Mapped SHAP feature attributions to **regulatory adverse-action reason codes**, making every rejected application explainable under lending compliance standards; deployed via FastAPI, Streamlit, Docker Compose, and GitHub Actions CI/CD

HMT-ECGNet – Lightweight Cardiac Triage Classifier

Python | PyTorch | FastAPI | Docker

github.com/MahboobAlam0/hmt_ecgnet

- Built a **338K-parameter** 1D CNN on PTB-XL (12-lead, multi-label ECG dataset) using Asymmetric Focal Loss for class imbalance – designed specifically to replace ResNet-1D (8.7M params) in CPU-constrained clinical settings; achieved **Macro-AUROC 0.92**, **MI AUROC 0.98**, **Macro-F1 0.73** at **26x fewer parameters** with **10-30 ms** CPU inference per ECG; served via Dockerized FastAPI